CLASS TEST-1

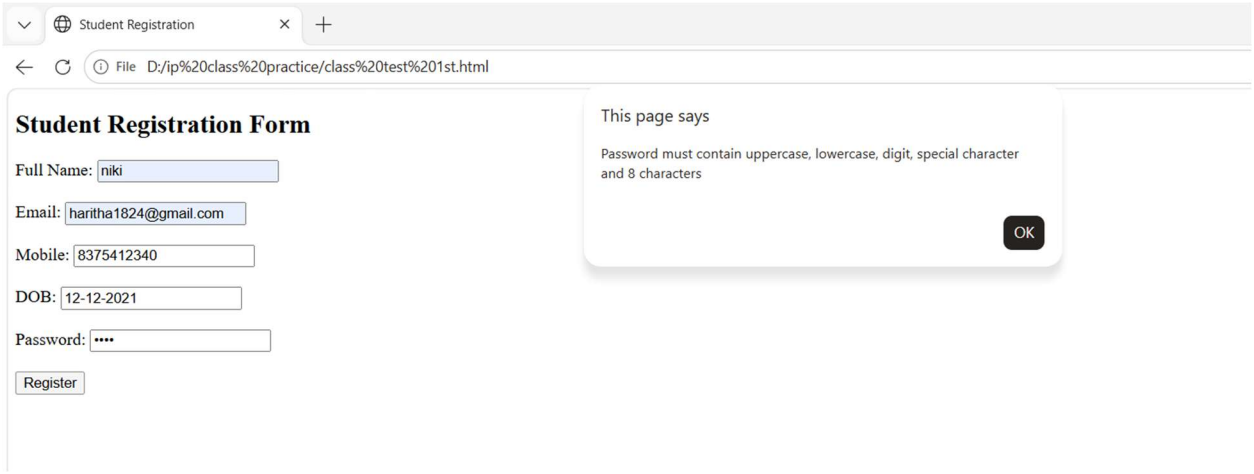
1. Student Registration Validation: Write a JavaScript program to validate a student registration form containing a full name, email address, 10-digit mobile number, date of birth in DD-MM-YYYY format, and password. The password must contain at least one uppercase letter, one lowercase letter, one digit, and one special character, with a minimum length of eight characters.

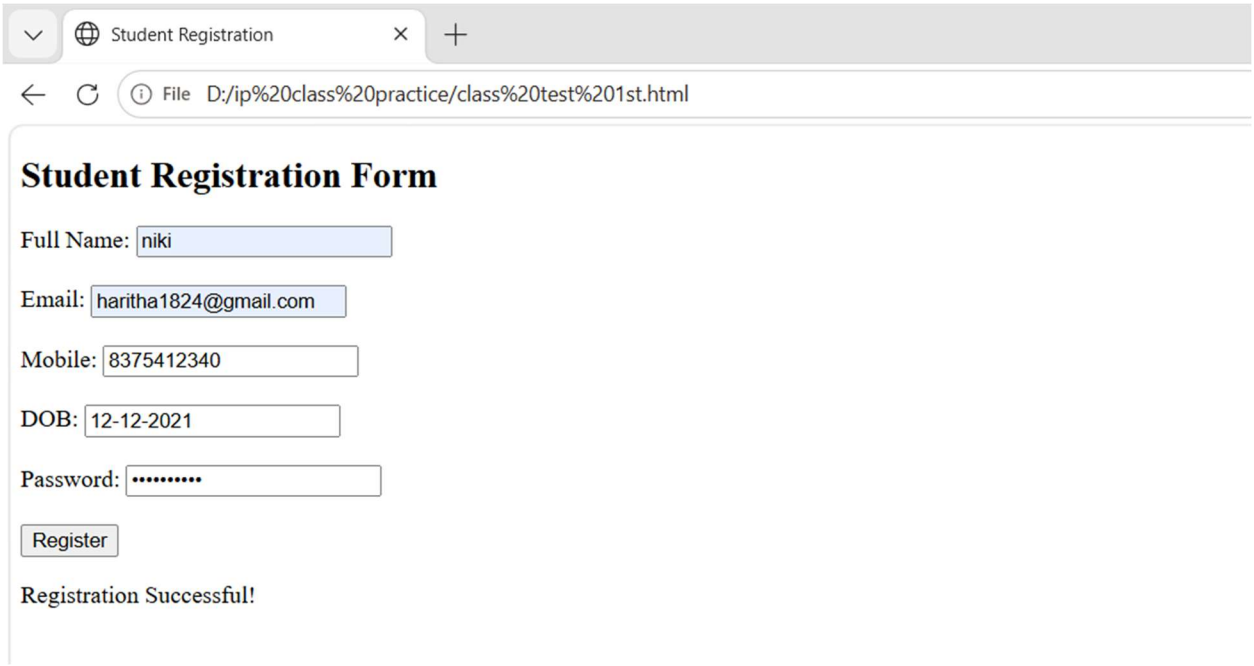
CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Student Registration</title>
</head><body>
<h2>Student Registration Form</h2>
<form onsubmit="return validate()">
   Full Name: <input type="text" id="name" required><br><br>
   Email: <input type="text" id="email" required><br><br>
   Mobile: <input type="text" id="mobile" required><br><br>
   DOB: <input type="text" id="dob" placeholder="DD-MM-YYYY" required><br><br>
   Password: <input type="password" id="pass" required><br><br>
   <button type="submit">Register</button>
</form>
<p id="result"></p>
<script>
function validate() {
   let n = document.getElementById("name").value;
   let e = document.getElementById("email").value;
   let m = document.getElementById("mobile").value;
```

```
    let d = document.getElementById("dob").value;
    let p = document.getElementById("pass").value;
    let nr = /^[A-Za-z ]+$/;
    let er = /^[^\s@]+@[^\s@]+\.[A-Za-z]{2,}$/;
    let mr = /^[0-9]{10}$/;
    let dr = /^\d{2}-\d{2}-\d{4}$/;
    let pr = /^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$!%*?&]).{8,}$/;
    if (!nr.test(n))
        alert("Invalid name");
    else if (!er.test(e))
        alert("Invalid email");
    else if (!mr.test(m))
        alert("Mobile must contain 10 digits");
    else if (!dr.test(d))
        alert("DOB must be DD-MM-YYYY");
    else if (!pr.test(p))
        alert("Password must contain uppercase, lowercase, digit, special character and 8
characters");
    else {
        document.getElementById("result").innerHTML =
            "Registration Successful!";
    }
    return false;
}
</script>
</body>
</html>
```

OUTPUT:





2. Password Strength Validation: Write a JavaScript program to validate a password with the following rules: minimum 10 characters, at least two digits, one uppercase letter, one lowercase letter, one special character, and no whitespace. Display an appropriate error message for each failed condition.

CODE:

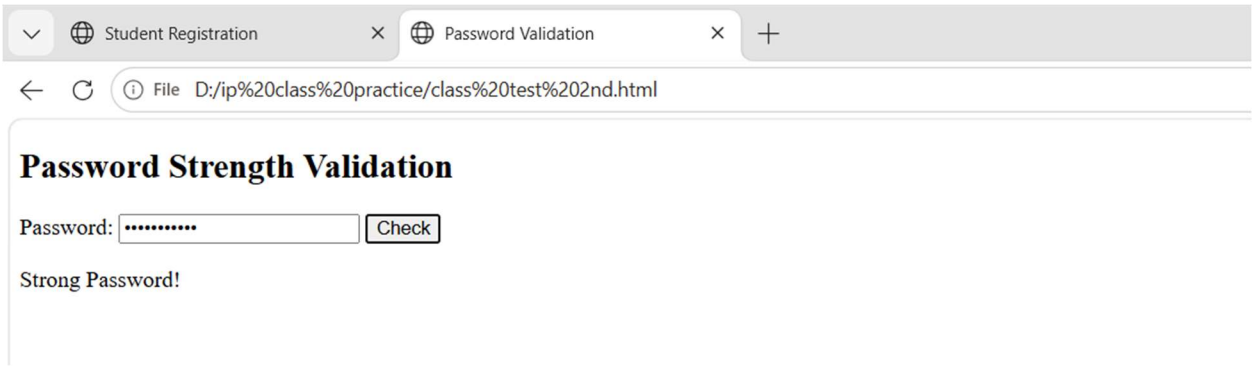
<!DOCTYPE html>

<html>

```
<head>
   <title>Password Validation</title>
</head>
<body>
<h2>Password Strength Validation</h2>
<form onsubmit="return validate()">
   Password: <input type="password" id="pass" required>
   <button type="submit">Check</button>
</form>
<p id="result"></p>
<script>
function validate() {
   let p = document.getElementById("pass").value;
   let errors = [];
   if (p.length < 10)
      errors.push("Minimum 10 characters required");
   if ((p.match(/\d/g) || []).length < 2)
      errors.push("At least two digits required");
   if (!/[A-Z]/.test(p))
      errors.push("One uppercase letter required");
   if (!/[a-z]/.test(p))
      errors.push("One lowercase letter required");
   if (!/[^A-Za-z0-9\s]/.test(p))
      errors.push("One special character required");
   if (/\s/.test(p))
      errors.push("Spaces are not allowed");
```

```
    document.getElementById("result").innerHTML =
        errors.length ? errors.join("<br>") : "Strong Password!";
    return false;
}
</script>
</body>
</html>
```

OUTPUT:



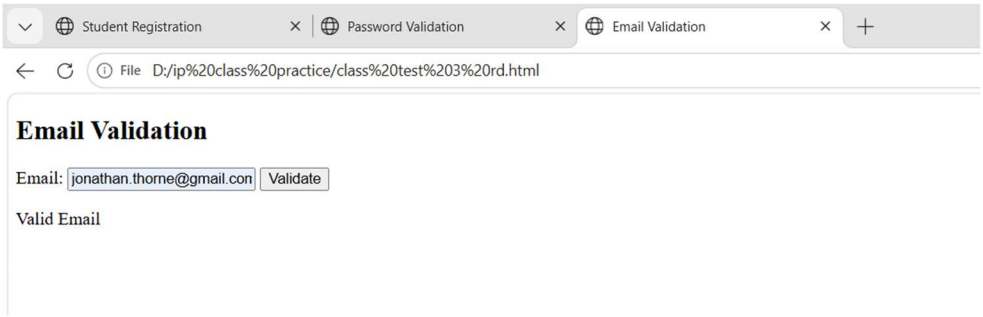
3. Email Address Validation: Write a JavaScript program to validate email addresses according to these rules: the address must contain exactly one @, a non-empty username, a domain name, and a valid-looking extension of at least two letters. Reject spaces and consecutive dots.

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Email Validation</title>
</head>
<body>
<h2>Email Validation</h2>
```

```
<form onsubmit="return validate()">
   Email: <input type="text" id="email" required>
   <button type="submit">Validate</button>
</form>
<p id="result"></p>
<script>
function validate() {
   let e = document.getElementById("email").value;
   let at = (e.match(/@/g) || []).length;
   let pattern =
      /^(?!.*\.\.)[A-Za-z0-9_%+-]+(?:\.[A-Za-z0-9_%+-]+)*@[A-Za-z0-9-]+(?:\.[A-Za-z0-9-]+)*\.[A-Za-z]{2,}$/;
   if (at === 1 && pattern.test(e))
      document.getElementById("result").innerHTML = "Valid Email";
   else
      document.getElementById("result").innerHTML = "Invalid Email";
   return false;
}
</script>
</body>
</html>
```

OUTPUT:



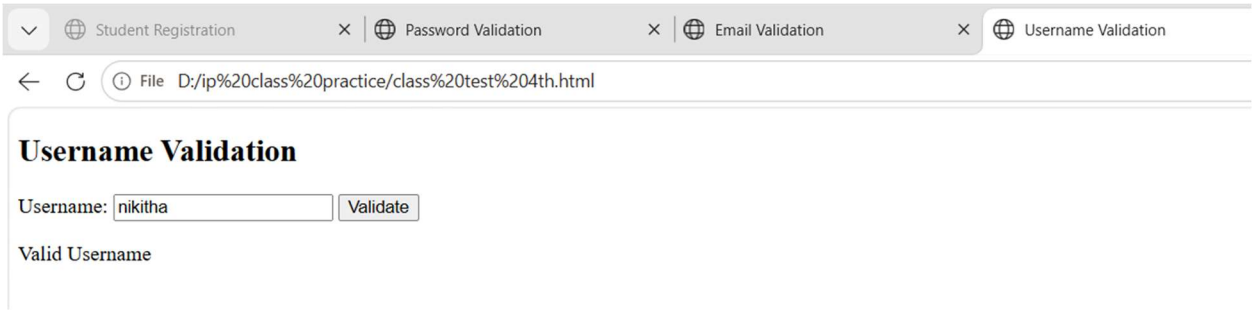
4. Username Validation: Write a JavaScript program to validate a username that must start with a letter, contain only letters, digits, and underscores, have a length of 6–15 characters, and must not end with an underscore or contain consecutive underscores.

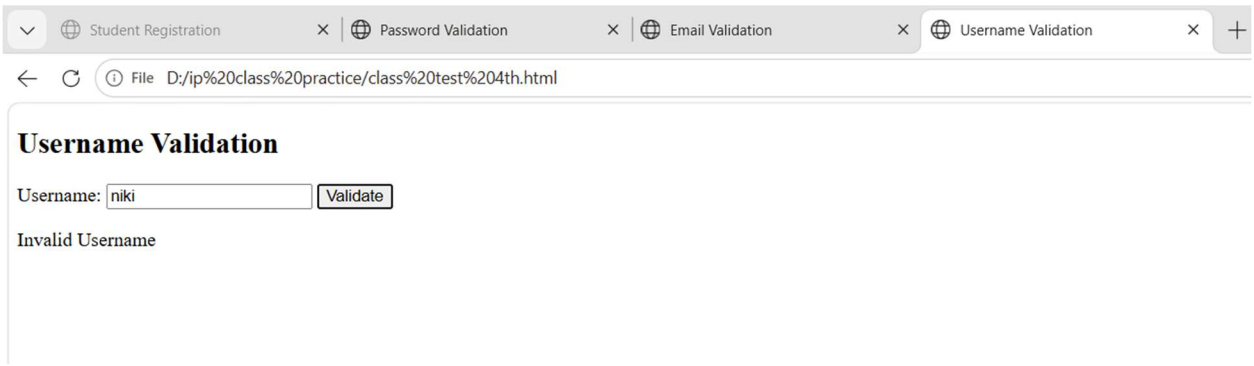
CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Username Validation</title>
</head>
<body>
<h2>Username Validation</h2>
<form onsubmit="return validate()">
   Username: <input type="text" id="user" required>
   <button type="submit">Validate</button>
</form>
<p id="result"></p>
<script>
function validate() {
   let u = document.getElementById("user").value;
   let pattern =
      /^(?=.{6,15}$)(?!.*__)[A-Za-z][A-Za-z0-9_]*[A-Za-z0-9]$/;
   if (pattern.test(u))
      document.getElementById("result").innerHTML = "Valid Username";
   else
      document.getElementById("result").innerHTML = "Invalid Username";
```

```
    return false;
}
</script>
</body>
</html>
```

OUTPUT:





5. Extract and Analyse Data: Write a JavaScript program to extract all mobile numbers, email addresses, and dates from a paragraph using regular expressions. Display each category separately.

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Extract Data</title>
</head>
<body>
```

```
<h2>Extract Data Using Regular Expressions</h2>

<textarea id="text" rows="5" cols="50">Contact Nikitha at 9876543210 or 9123456780. Email nikitha@gmail.com and student@yahoo.in. Dates: 24-10-2005 and 15-08-2026.</textarea>

<br><br>

<button onclick="extract()">Extract Data</button>

<h3>Mobile Numbers</h3>

<p id="mobile"></p>

<h3>Email Addresses</h3>

<p id="email"></p>

<h3>Dates</h3>

<p id="date"></p>

<script>

function extract() {

    let t = document.getElementById("text").value;

    let m = t.match(/\b[6-9]\d{9}\b/g) || [];

    let e = t.match(/\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b/g) || [];

    let d = t.match(/\b\d{2}-\d{2}-\d{4}\b/g) || [];

    document.getElementById("mobile").innerHTML = m.join("<br>") || "None found";

    document.getElementById("email").innerHTML = e.join("<br>") || "None found";

    document.getElementById("date").innerHTML = d.join("<br>") || "None found";

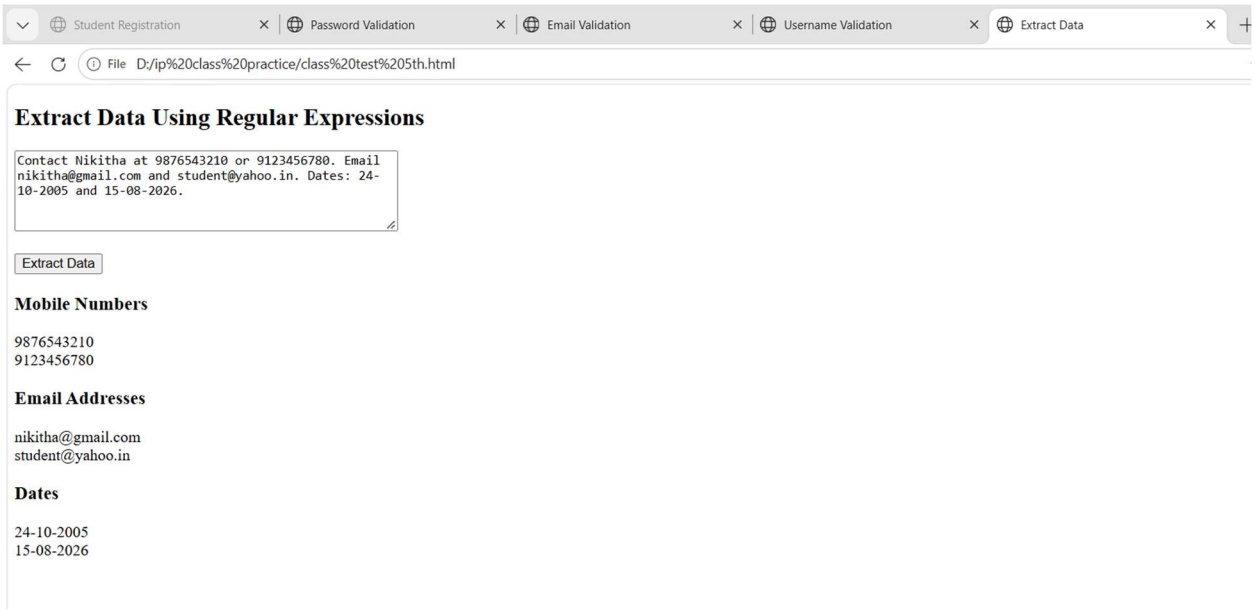
}

</script>

</body>

</html>
```

OUTPUT:


Student Registration
Password Validation
Email Validation
Username Validation
Extract Data
File D:/ip%20class%20practice/class%20test%205th.html
Extract Data Using Regular Expressions
Contact Nikitha at 9876543210 or 9123456780. Email nikitha@gmail.com and student@yahoo.in. Dates: 24-10-2005 and 15-08-2026.
Extract Data
Mobile Numbers
9876543210
9123456780
Email Addresses
nikitha@gmail.com
student@yahoo.in
Dates
24-10-2005
15-08-2026